Lesson 3: How High Can You Go?

Unit

Energy: Stores and Transfers

Lesson focus

Students meet gravitational field strength as "newtons per kilogram", calculate the energy in the gravitational potential store with $E_p = m\ g\ h$, and rearrange it to find a height.

Specification

AQA GCSE Physics 4.1.1.2 Changes in energy. The equation is on the AQA Physics equation sheet, and g is always given in calculations. This unit uses g = 9.8 N/kg throughout.

Scope

Students calculate E_p, and rearrange to find h or m.

Weight (W = m g) is taught later, in the Forces unit. This lesson only needs the idea that gravity pulls each kilogram down with 9.8 N. That one sentence is reused in Lesson 4.

Learning objectives

By the end of the lesson, students should be able to:

1. Explain what the gravitational field strength g means.
2. Calculate the energy in the gravitational potential store using $E_p = m\ g\ h$.
3. Use the change in height, not the height above the ground, when an object moves between two points.
4. Rearrange the equation to find a height or a mass.

Key vocabulary

Keyword	Student-friendly definition
Gravitational potential store	The energy an object has because it is raised in a gravitational field
Gravitational field strength (g)	The force of gravity on each kilogram; 9.8 N/kg on Earth
Newton (N)	The unit of force
Height (h)	The vertical distance an object is raised, in metres (m)

The equation

Quantity	Symbol	Unit
Gravitational potential energy	E_p	joule (J)
Mass	m	kilogram (kg)
Gravitational field strength	g	newton per kilogram (N/kg)
Height	h	metre (m)

$$E_p = m\ g\ h$$

gravitational potential energy = mass × gravitational field strength × height

What g means

> *On Earth, gravity pulls every kilogram down with a force of 9.8 newtons. That is what 9.8 N/kg means.*

Hand round a 1 kg mass. That pull is 9.8 N. On the Moon it would be 1.6 N; on Jupiter, about 25 N. So lifting the same object through the same height stores less energy on the Moon.

Do not introduce W = m g as an equation. Students will meet it in Forces.

Equipment and resources

- A 1 kg mass; a set of everyday objects with masses labelled
- Metre rules or a tape measure
- Calculators

Starter: Which stores more?

Rank by energy in the gravitational potential store:

1. A 1 kg bag of sugar on a 1 m table
2. The same bag on a 2 m shelf
3. A 2 kg bag on the 1 m table
4. The 1 kg bag on a 1 m table on the Moon

Students should reach: 2 and 3 are equal and largest; 1 next; 4 smallest. Ask what the three things are that matter (mass, height, strength of gravity).

Main teaching: Calculating E_p

Write the equation, substitute, calculate.

Worked example 1:

> *A 2.0 kg book is lifted onto a shelf 1.5 m high. Calculate the energy transferred to its gravitational potential store. ($g = 9.8$ N/kg)*
> *$E_p = m\ g\ h$*
> *$E_p = 2.0 \times 9.8 \times 1.5$*
> *$E_p = 29.4$ J*

Worked example 2 (change in height):

> *A 55 kg hiker climbs from a car park 120 m above sea level to a summit 420 m above sea level. Calculate the increase in the energy in their gravitational potential store.*
> *$h = 420 - 120 = 300$ m*
> *$E_p = m\ g\ h$*
> *$E_p = 55 \times 9.8 \times 300$*
> *$E_p = 161\,700$ J*

Point out that the car park's height above sea level does not matter; only the height gained does.

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Main teaching: Finding the height

Rearrange in symbols first, then substitute.

> *$E_p = m\ g\ h$*
> *Divide both sides by m g:*
> *$h = E_p \div (m\ g)$*
> *In the same way, $m = E_p \div (g\ h)$*

Worked example 3:

> *A 60 kg climber gains 17 640 J in their gravitational potential store. How high did they climb?*
> *$h = E_p \div (m\ g)$*
> *$h = 17\ 640 \div (60 \times 9.8)$*
> *$h = 17\ 640 \div 588$*
> *$h = 30$ m*

Common error: dividing by 60 and then multiplying by 9.8. Teach students to work out m g first, or to use brackets on the calculator.

Activity 1: Calculation page

Workbook page in three sections:

1. Straight substitution (masses in kg, then in g).
2. Change in height (stairs, lifts, a ski slope).
3. Finding h or m, including one in kilojoules (convert to joules first).

Activity 2: Lift it

Pairs choose three objects, measure the mass and the height of a shelf or bench, and calculate the energy transferred when each object is lifted onto it. Then: how high would you have to lift the lightest object for it to store the same energy as the heaviest? (Uses $h = E_p \div (m\ g)$.)

Check for understanding

1. What does g = 9.8 N/kg mean?
2. Calculate E_p for a 5.0 kg bag lifted 2.0 m. (98 J)
3. Why would the same lift store less energy on the Moon?
4. A 0.50 kg ball gains 24.5 J. How high was it lifted? (5.0 m)
5. A crate lifted 2.0 m gains 392 J. What is its mass? (20 kg)

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Plenary

Show a photograph of a reservoir behind a dam. Ask: where is the energy stored, and what would you need to know to calculate it? (The mass of water and its height above the turbines.) Link forward: in Lesson 6, that energy is transferred to the kinetic store as the water falls.

Possible prep

Your school bag has a mass of about 5 kg. Calculate the energy transferred to its gravitational potential store when you carry it up one flight of stairs (about 3 m). Then calculate it for the whole school day's stair climbing.

Teacher notes

Common mistakes:

1. Using height above the ground instead of the change in height.
2. Using 10 N/kg out of habit. AQA gives 9.8 N/kg; use it.
3. Mass in grams.
4. In rearranging, dividing by only one of m and g.

Mass and weight are often confused. Students may say "the book weighs 2 kg". Do not make this the focus of the lesson, but model the correct language: "its mass is 2 kg; gravity pulls it down with 19.6 N".

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